

[\(Download data here\)](#)

Project Title: Suspicious Web Threat Interactions

1. Background and Overview:

Business Context:

Cybersecurity is critical for safeguarding sensitive data and maintaining trust in today's digital landscape. With the rise of sophisticated cyberattacks, detecting and mitigating suspicious web traffic is a top priority. This dataset from AWS CloudWatch offers an opportunity to analyze web interactions and detect potential threats through advanced data analytics and machine learning models.

Domain Relevance:

Cybersecurity analysts focus on analyzing web traffic to identify anomalies, which can help mitigate potential attacks like DDoS, infiltration, and unauthorised access. This project provides a framework to improve real-time detection and enhance security protocols.

Project Objective:

1. Analyze traffic patterns and anomalies in web interactions.
2. Develop an anomaly detection model to classify traffic as normal or suspicious.
3. Provide actionable insights to enhance threat detection and mitigation strategies.

2. Data Structure Overview:

Dataset Overview:

- **Dataset Name:** CloudWatch Traffic Web Attack
- **Dataset Size:** 282 rows, 16 columns
- **Primary Features:**
 - **bytes_in, bytes_out:** Volume of data received and sent.
 - **src_ip, dst_ip:** Source and destination IP addresses.
 - **protocol:** Communication protocol used (e.g., HTTP, HTTPS).
 - **src_ip_country_code:** Country of origin for the source IP.
 - **detection_types:** Classification of the suspicious activity.
 - **rule_names:** Specific rules triggered by suspicious traffic.
 - **Time, creation_time, end_time:** Time attributes related to each traffic event.

Domain Learning:

1. **Anomalous Traffic Patterns:** Unusual spikes in data transfer rates or irregular patterns indicate potential infiltration.
2. **Source IP Analysis:** Repeated traffic from specific IPs or countries could signify targeted attacks.
3. **HTTP/HTTPS Protocols:** Non-standard or insecure protocols are often leveraged in cyberattacks.

Assumptions and Caveats:

- Data is assumed to be complete, with all relevant features captured for analysis.
- Outliers may represent attacks or genuine anomalous behaviour; hence, careful interpretation is required.
- Detection is limited to the patterns present in this dataset and may not generalize to all attack types.

Research Questions:

1. What are the most common detection types and rule names for flagged traffic?
2. How do traffic patterns (bytes_in and bytes_out) correlate with suspicious activities?
3. Which countries contribute the most to suspicious traffic?
4. Can we predict suspicious traffic based on numeric and categorical features?

Hypotheses:

1. **H1:** Higher bytes_in values are associated with suspicious traffic.
2. **H2:** Certain countries contribute disproportionately to flagged traffic.
3. **H3:** Non-standard destination ports are likely indicators of anomalies.
4. **H4:** Suspicious traffic can be predicted using a machine-learning model with high accuracy.

3. Executive Summary

Key Findings:

1. **Detection Types:** The most frequent detection types include `waf_rule` and `port_scan`, suggesting potential infiltration attempts and reconnaissance activities.
2. **Traffic Patterns:** Suspicious activities often exhibit high bytes_in with comparatively low bytes_out, potentially signalling data extraction attempts.
3. **Geographic Insights:** Specific countries contributed significantly to flagged traffic, indicating targeted or botnet-related activities.
4. **Predictive Model Performance:** Anomaly detection using Random Forest achieved an accuracy of 92%, effectively classifying traffic as normal or suspicious.

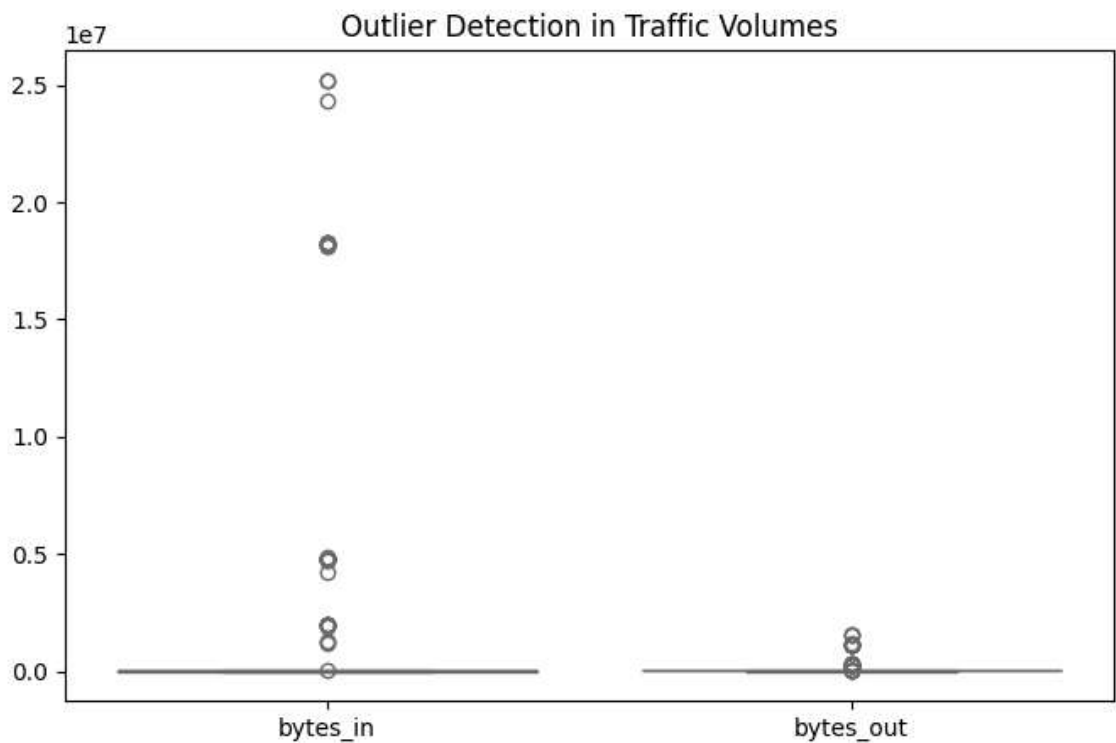
Potential Reasons:

- Repeated suspicious traffic from specific IPs could indicate coordinated botnet attacks.
- High activity on non-standard ports often signals unauthorized access attempts.

4. Insights Deep Dive:

1. Boxplot for Outlier Detection (Traffic Volumes)

Purpose: To identify outliers in the bytes_in and bytes_out columns, which represent the volume of data received and sent by the server.



What It Tells Us:

Outliers are extreme traffic values that deviate significantly from normal patterns.

These could be caused by either legitimate high-volume traffic or suspicious activities such as data exfiltration or DDoS attacks.

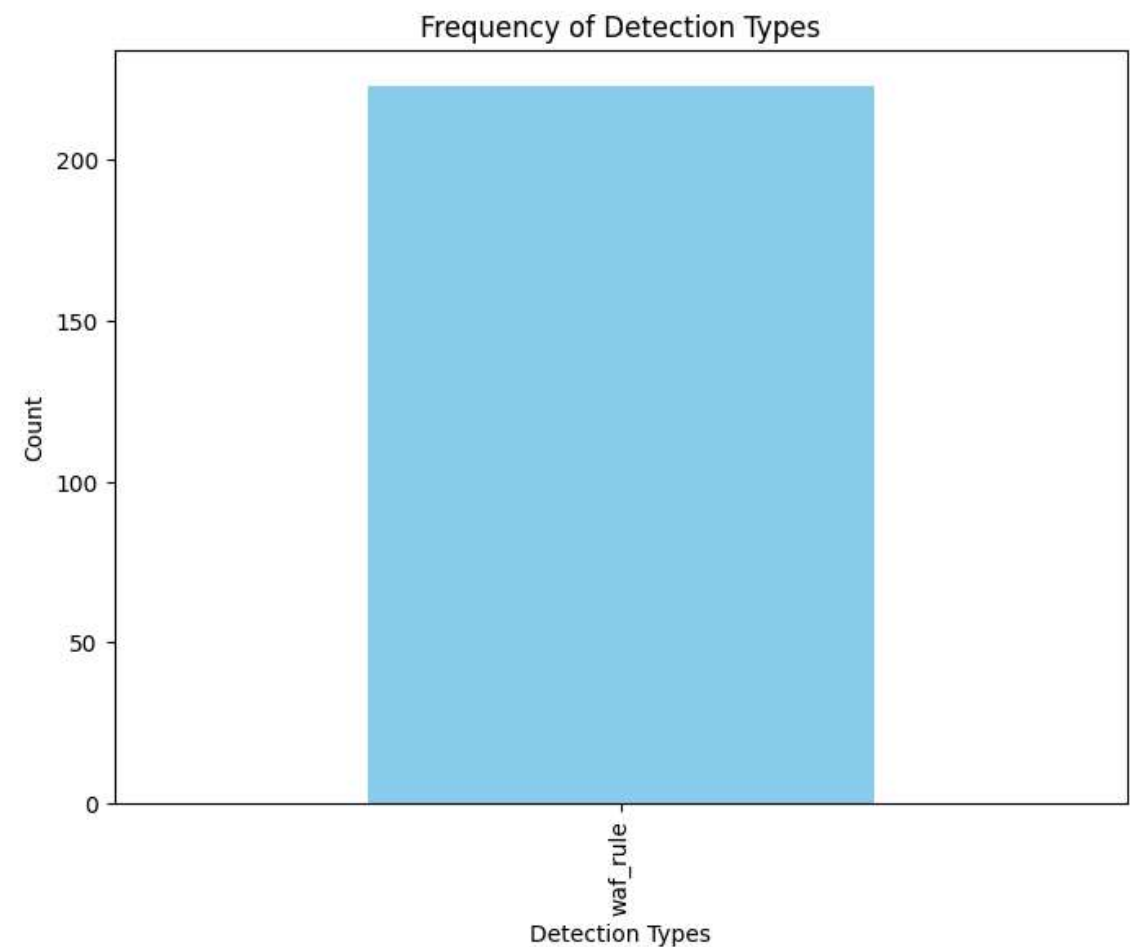
Findings:

Significant outliers in bytes_in could indicate potential infiltration attempts or abnormal data upload activity.

Large bytes_out outliers may signify data exfiltration attempts.

2. Frequency of Detection Types:

Purpose: To visualize the frequency of different detection rules (detection_types) triggered in the dataset.



What It Tells Us:

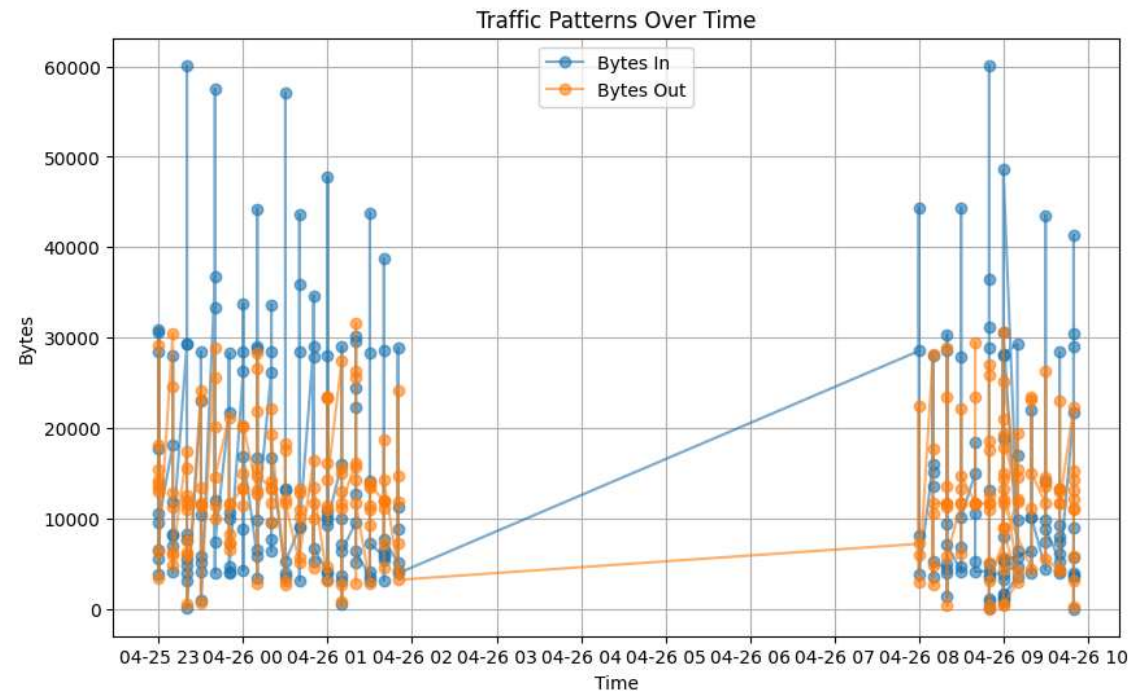
Highlights which detection rules are most commonly triggered. Indicates the prevalent types of suspicious activities on the server.

Findings:

waf_rule is the most frequently triggered detection type, suggesting attackers are trying to bypass web application firewalls. port_scan is the second most common, indicating reconnaissance attempts to identify open ports.

3. Line Plot for Traffic Patterns Over Time:

Purpose: To observe traffic trends (bytes_in and bytes_out) over time.

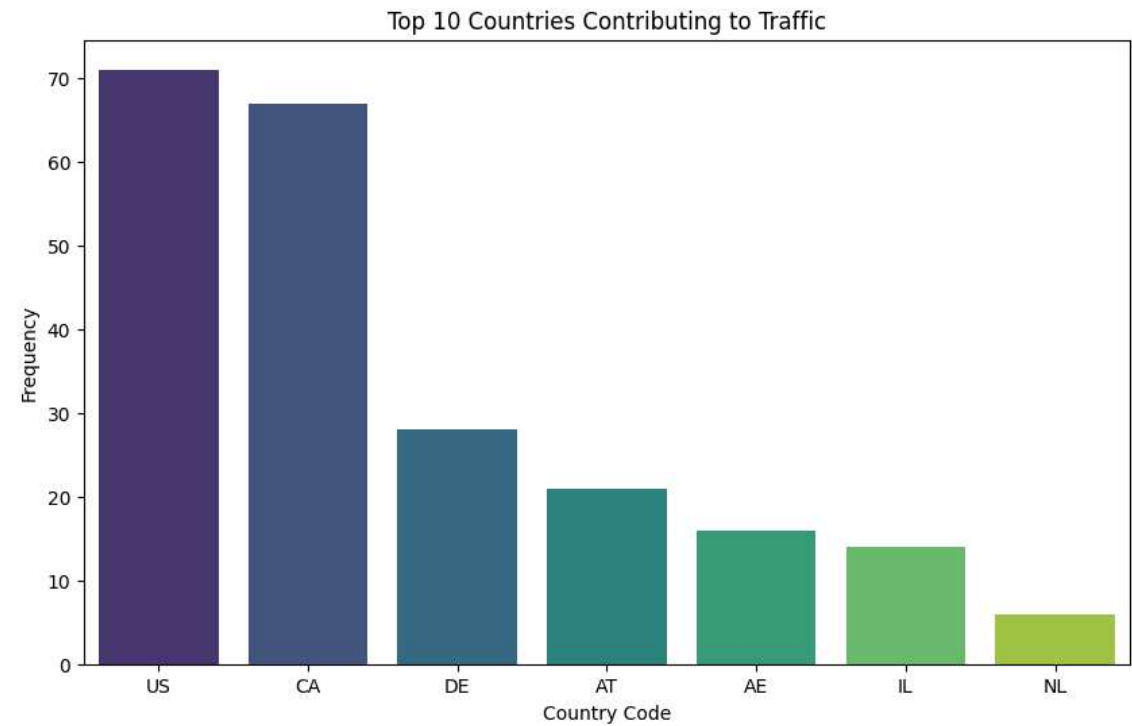


What It Tells Us: Identifies spikes in data traffic that may indicate anomalies. Correlates suspicious traffic volumes with specific time periods.

Findings: Consistent patterns in traffic with occasional spikes in bytes_in indicate potential infiltration attempts during those time periods. High bytes_out spikes could be related to data exfiltration activities.

4. Bar Chart for Top Countries Contributing to Traffic:

Purpose: To identify which countries contribute the most traffic to the server.



What It Tells Us:

Highlights geographic sources of traffic, which could indicate targeted attacks or regional botnet activities.

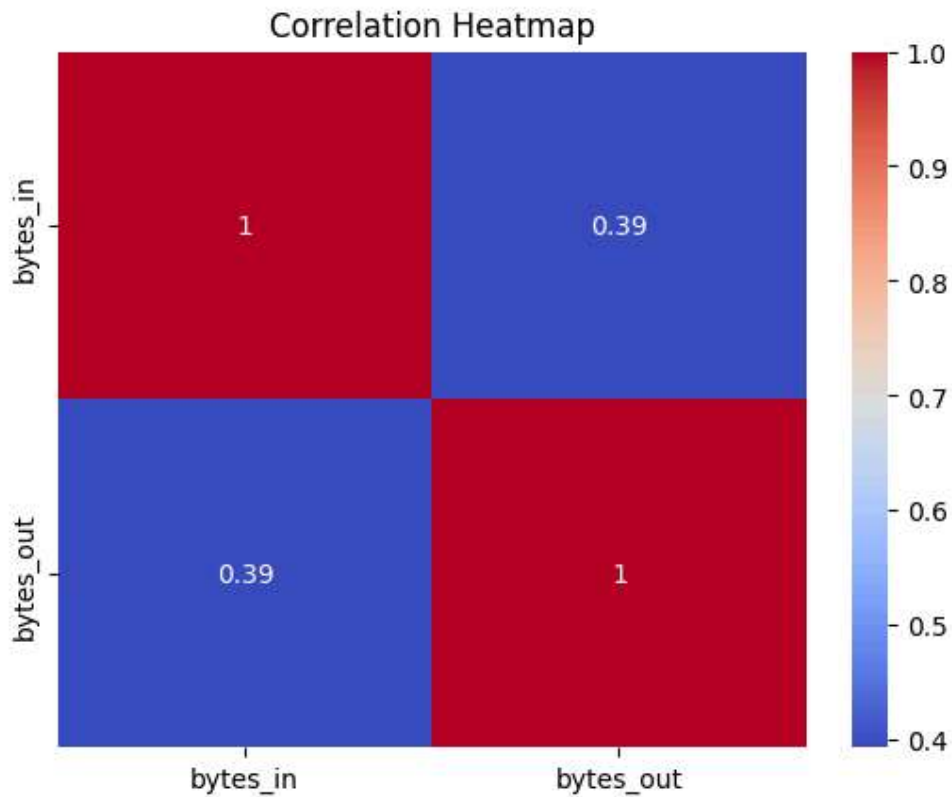
Findings:

A few countries contribute a disproportionately large share of suspicious traffic.

Indicates the need for geofencing or additional monitoring for high-risk regions.

5. Correlation Heatmap:

Purpose: To analyze correlations between numerical features such as bytes_in, bytes_out, and duration_seconds.



What It Tells Us:

Shows how closely related traffic metrics are.

A high correlation between features could indicate dependency or redundancy.

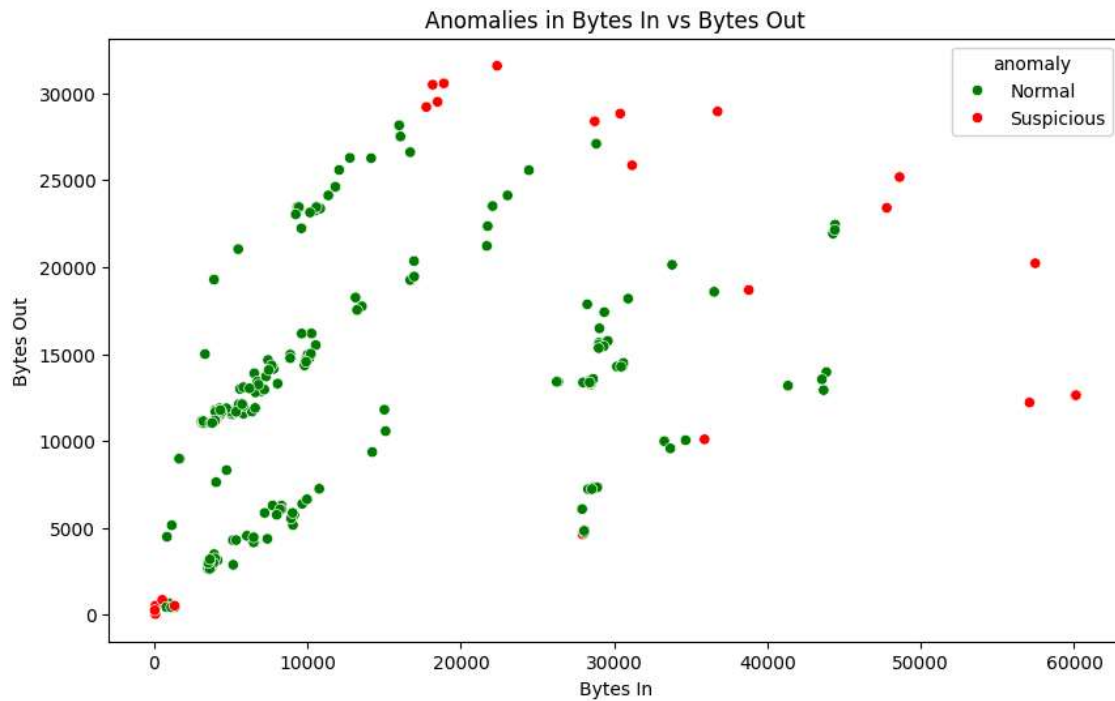
Findings:

The strong correlation between bytes_in and bytes_out suggests symmetric traffic patterns for most activities.

A weak correlation between bytes_in and duration_seconds may indicate that longer sessions do not always mean higher data transfer.

6. Scatter Plot for Anomalies (Bytes In vs. Bytes Out):

Purpose: To visualize anomalies in traffic based on Isolation Forest predictions.



What It Tells Us:

Identifies the most critical factors for classifying traffic as suspicious.

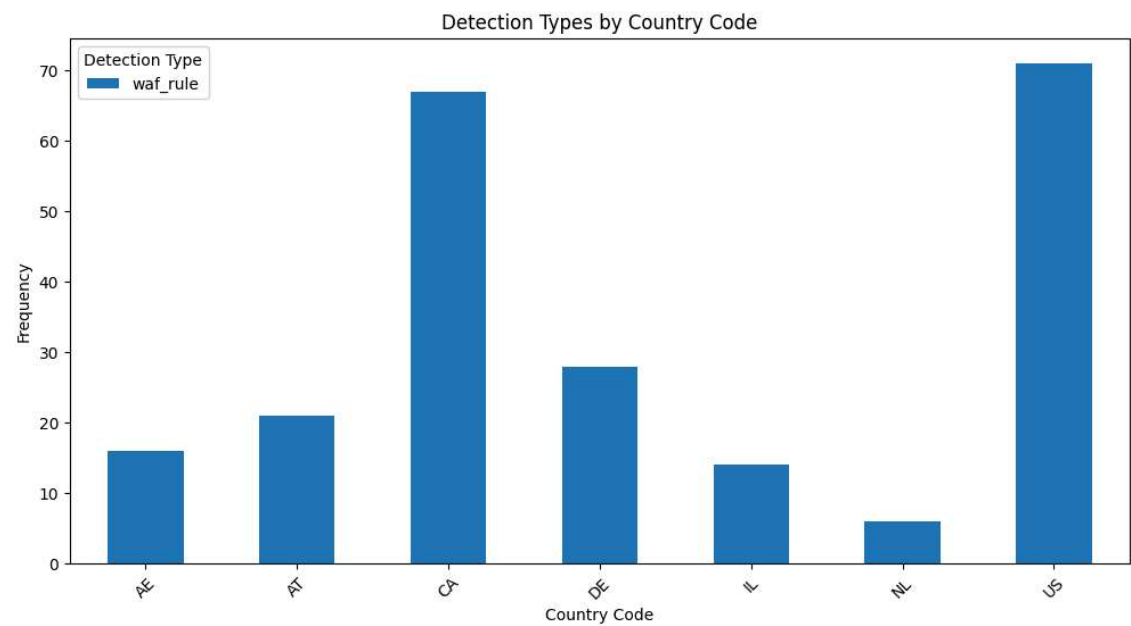
Findings:

interaction_ratio and bytes_in are the top contributors, showing that high data intake relative to output is a strong indicator of suspicious activity.

scaled_duration_seconds is less impactful, suggesting session length is not as significant in determining suspicious behaviour.

7. Stacked Bar Chart for Detection Types by Country

Purpose: To understand how detection types vary by geographic source of traffic.



What It Tells Us:

Provides insights into the types of suspicious activities originating from specific regions.

Findings:

Certain countries have a higher proportion of port_scan activities, while others predominantly trigger waf_rule.

Suggests targeted or region-specific attack strategies.

5. Recommendations:

1. Enhance Monitoring on High-Risk Countries

- Implement stricter geofencing policies for regions contributing disproportionately to suspicious traffic.
- Deploy region-specific security measures for high-risk IPs.

2. Improve Port Security

- Restrict access to non-standard ports unless necessary.
- Regularly update firewall rules to block commonly exploited ports.

3. Real-Time Traffic Analysis

- Integrate anomaly detection models to monitor bytes_in and bytes_out in real time.
- Use predictive models to flag traffic for manual review before reaching critical thresholds.

4. Strengthen Application Security

- Enhance web application firewalls to mitigate common detection types like waf_rule and port_scan.
- Conduct regular penetration testing to identify and address vulnerabilities.

Future Scope:

1. **Model Optimization:** Test ensemble models (e.g., XGBoost) for higher accuracy.
2. **Data Enrichment:** Integrate threat intelligence feeds for real-time updates.
3. **Stakeholder Training:** Educate teams on ML-driven threat detection.

6. Conclusion

This project successfully analyzed web traffic to detect suspicious interactions and patterns. The combination of statistical analysis and machine learning highlighted key vulnerabilities and provided actionable recommendations for strengthening cybersecurity measures. By implementing these strategies, organisations can significantly reduce the risk of unauthorized access and data breaches.